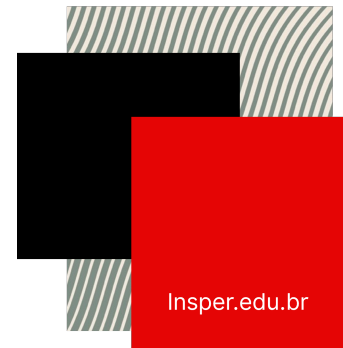
Economics of Organizations and Incentives

MPE – 4th term, 2026

Lecture 3: Mixed strategies and dynamic games



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Recap: Nash equilibrium

- A **best response** to s_{-i} is a strategy that maximizes $u_i(\cdot, s_{-i})$.
- A **Nash equilibrium** is a profile of mutual best responses: no player gains from a unilateral deviation.
- Cournot: $q_1^* = q_2^* = \frac{2}{3}q^m$. Bertrand: $p_1^* = p_2^* = c$.

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But some games have no Nash equilibrium in pure strategies. Think of matching pennies, or of a penalty kick: each player wants to be unpredictable.

Bertrand with capacity constraints

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What is the Nash equilibrium if firms have **limited capacity**?

Numerical example: demand $Q(p) = 100 - p$; capacity $\bar{q} = 60$ per firm; marginal cost $c = 5$.

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Profit of firm 1 (the cheaper firm sells first; the other serves the residual demand):

$$\Pi_1 = \begin{cases} (p_1 - 5)\,\min(100 - p_1,\ 60) & \text{if } p_1 < p_2, \\ \dfrac{(p_1 - 5)(100 - p_1)}{2} & \text{if } p_1 = p_2, \\ 0 & \text{if } p_1 > p_2 \text{ and } p_2 \geq 40, \\ (p_1 - 5)\,(100 - p_1 - 60) & \text{if } p_1 > p_2 \text{ and } p_2 < 40. \end{cases}$$

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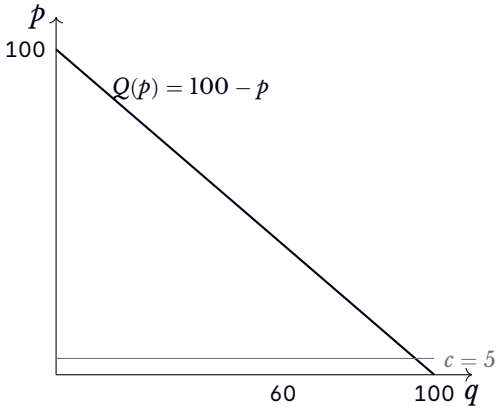
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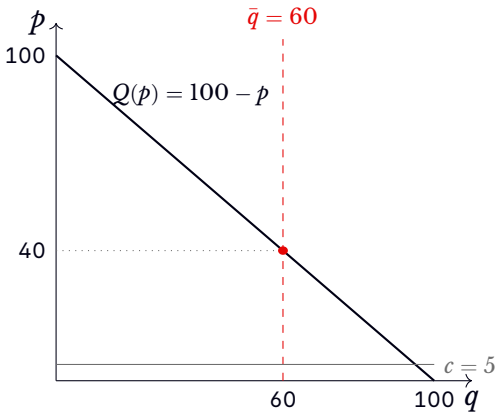
Edgeworth paradox: with capacity constraints, Bertrand has no pure-strategy Nash equilibrium (general model in the appendix).

Demand and capacity



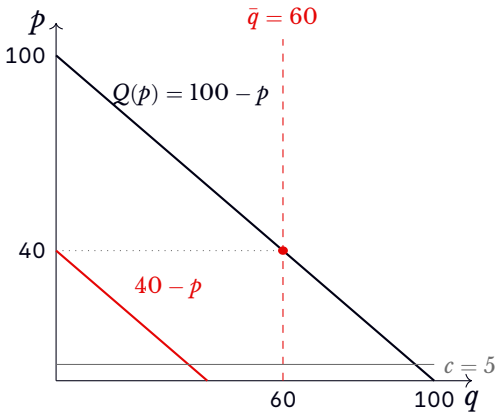
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- A single firm can serve at most 60 units. At prices below 40, demand exceeds one firm's capacity.
- If the cheaper firm sells 60, the more expensive firm faces the **residual demand** $100 - p - 60 = 40 - p$.

No equilibrium in pure strategies

Case 1: $p_i = p_j = c = 5$

If i raises its price to 6, it serves the residual demand $94 - 60 = 34$. Profit $= 1 \times 34 = 34 > 0$. Profitable deviation.

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Case 2: $p_i = p_j = 10 > c$

Each firm sells 45 and earns $5 \times 45 = 225$. If i cuts to 9.50, it sells its full capacity of 60 and earns $4.5 \times 60 = 270 > 225$. Profitable deviation.

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Case 4: $p_i > p_j$**, with** $p_j < 40$

Demand at p_j exceeds 60, so j sells its full capacity. Raising its price slightly, still below p_i, j keeps selling 60 at a higher margin. Profitable deviation.

Mixed strategies

Mixed strategies

Definition

A **mixed strategy** of player i is a probability distribution over S_i: $\sigma_i : S_i \to [0,1]$ such that $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$. We denote the set of mixed strategies by $\Sigma_i := \Delta(S_i)$, and a mixed-strategy profile by $\sigma = (\sigma_1, \ldots, \sigma_n) \in \Sigma := \Sigma_1 \times \cdots \times \Sigma_n$.

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- A player randomizes to avoid being predictable: to surprise, or to avoid being surprised.
- Every pure strategy is a (degenerate) mixed strategy.
- With mixed strategies, players evaluate *expected* payoffs.

Mixed strategies: expected payoffs

Payoff of i given a profile σ:

$$u_i(\sigma) = \sum_{s \in S} \left[\sigma_1(s_1) \cdots \sigma_n(s_n)\right] u_i(s).$$

Mixed strategies: expected payoffs

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Payoff of i from the pure strategy s_i, given σ_{-i}:

$$u_i(s_i, \sigma_{-i}) = \sum_{s_{-i} \in S_{-i}} \left[\prod_{j \neq i} \sigma_j(s_j)\right] u_i(s_i, s_{-i}).$$

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A **Nash equilibrium** is a profile $\sigma^* \in \Sigma$ such that $u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*)$ for every $\sigma_i \in \Sigma_i$ and every $i \in I$.

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Principle of indifference

$\sigma^* \in \Sigma$ is a Nash equilibrium if and only if, for every $i \in I$,

1. $u_i(\sigma^*) = u_i(s_i, \sigma_{-i}^*)$ for every $s_i \in \operatorname{supp} \sigma_i^*$;
2. $u_i(\sigma^*) \geq u_i(s_i, \sigma_{-i}^*)$ for every $s_i \in S_i \setminus \operatorname{supp} \sigma_i^*$,

where $\operatorname{supp} \sigma_i := \{s_i \in S_i : \sigma_i(s_i) > 0\}$ is the support of σ_i.

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where $\operatorname{supp} \sigma_i := \{s_i \in S_i : \sigma_i(s_i) > 0\}$ is the support of σ_i.

In words: every strategy played with positive probability yields the same expected payoff, and no strategy outside the support yields more.

Principle of indifference: proof

[Indifference $\Rightarrow$ Nash] If σ^* satisfies (1) and (2) for every i, then for every $\sigma_i \in \Sigma_i$,

$$u_i(\sigma_i, \sigma^*_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(s_i, \sigma^*_{-i}) \;\leq\; \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(\sigma^*_i, \sigma^*_{-i}) = u_i(\sigma^*_i, \sigma^*_{-i}).$$

Principle of indifference: proof

[Nash $\Rightarrow$ indifference] Suppose σ^* is a Nash equilibrium but (1) fails: there are $s_j', s_j'' \in \operatorname{supp} \sigma_j^*$ with $u_j(s_j'', \sigma_{-j}^*) > u_j(s_j', \sigma_{-j}^*)$. Consider

$$\hat{\sigma}_j(s_j) = \begin{cases} \sigma_j^*(s_j) & \text{if } s_j \notin \{s_j', s_j''\}, \\ 0 & \text{if } s_j = s_j', \\ \sigma_j^*(s_j') + \sigma_j^*(s_j'') & \text{if } s_j = s_j''. \end{cases}$$

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Moving the probability of s_j' to s_j'' strictly raises the expected payoff, $u_j(\hat{\sigma}_j, \sigma_{-j}^*) > u_j(\sigma_j^*, \sigma_{-j}^*)$: a contradiction. Condition (2) follows in the same way. $\square$

Example: battle of the sexes in mixed strategies

		Chris: O	Chris: F
Alex	O	$2,\ 1$	$0,\ 0$
	F	$0,\ 0$	$1,\ 2$

Let Chris play O with probability $q \in [0,1]$. Alex's expected payoff from each action:

$$u_A(O,q) = 2q + 0\,(1-q) = 2q, \qquad u_A(F,q) = 0\,q + 1\,(1-q) = 1-q.$$

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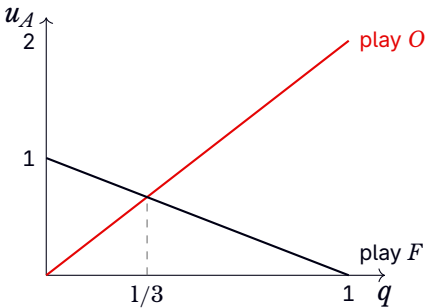
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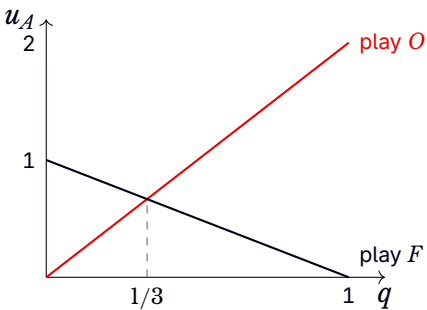
Alex prefers O if $2q > 1-q$, that is, if $q > \frac{1}{3}$.

Battle of the sexes: Alex's best response

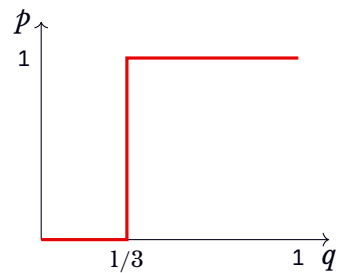


Expected payoff of each action of Alex

Battle of the sexes: Alex's best response



Expected payoff of each action of Alex



Alex's best response: probability p of O

$BR_A(q)$: F ($p = 0$) if $q < \frac{1}{3}$; any $p \in [0, 1]$ if $q = \frac{1}{3}$; O ($p = 1$) if $q > \frac{1}{3}$.

Insper

Battle of the sexes: Chris's best response

Let Alex play O with probability $p \in [0,1]$. Chris's expected payoff from each action:

$$u_C(O,p) = 1 \cdot p + 0\,(1-p) = p, \qquad u_C(F,p) = 0 \cdot p + 2\,(1-p) = 2 - 2p.$$

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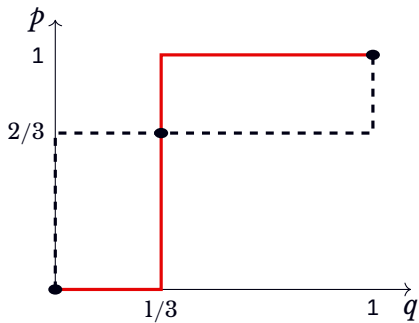
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Chris prefers O if $p > 2 - 2p$, that is, if $p > \frac{2}{3}$:

$$BR_C(p) = \begin{cases} q = 0 \ (F) & \text{if } p < \frac{2}{3}, \\ q \in [0,1] & \text{if } p = \frac{2}{3}, \\ q = 1 \ (O) & \text{if } p > \frac{2}{3}. \end{cases}$$

Insper

Battle of the sexes: all Nash equilibria

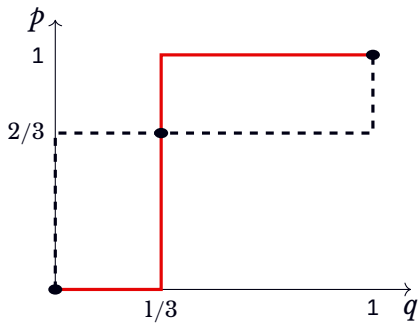


Best responses of Alex (solid) and Chris (dashed)

Equilibria = intersections of the best responses:

- $(p, q) = (0, 0)$: (F, F);
- $(p, q) = (1, 1)$: (O, O);
- $(p, q) = (\frac{2}{3}, \frac{1}{3})$: mixed equilibrium.

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In the mixed equilibrium, each probability is pinned down by the *other* player's payoffs; each player expects $2/3$, less than in either pure equilibrium.

Interpretations and existence

Common interpretations of mixed strategies:

- **Beliefs:** σ_{-i} describes what player i believes the others will do.
- **Empirical frequencies:** long-run frequencies of actions in repeated interactions.
- **Population distribution:** the share of a population of players using each action.

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Theorem (Nash, 1950)

Every finite normal-form game has a Nash equilibrium in mixed strategies.

Note: a profile of pure strategies is also a profile of mixed strategies (why?).

Dynamic games

Sequential battle of the sexes

Imagine that Alex finishes work at 3:00 p.m., while Chris finishes work at 5:00 p.m. This gives Alex ample time to get to either the football game or the opera and then to call Chris at 4:45 p.m. and announce “I am here”. Chris then has to choose where to go. Where should Chris go?

Sequential battle of the sexes

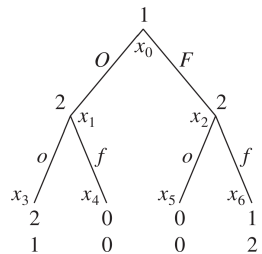
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Insper

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- every node x that is not terminal is assigned either to a player $i(x)$, with action set $A_i(x)$, or to Nature.

The extensive-form game

Games that unfold over time. Elements:

1. the set of players, N;
2. the players' payoffs as a function of outcomes;
3. a game tree, with the information each player has when called to move.

The structure of the extensive-form game is common knowledge among all the players.

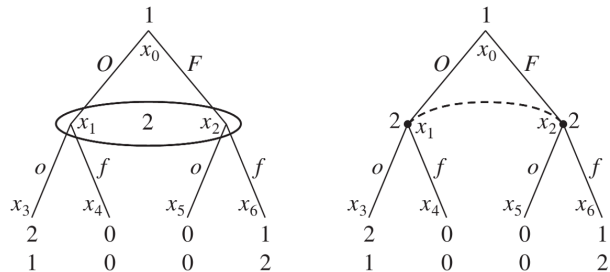
Information sets

Every player i has a collection of **information sets** $h_i \in H_i$ that partition the nodes at which player i moves, with the following properties:

1. if h_i is a singleton that includes only x, then player i, who moves at x, knows that she is at x;
2. if $x \neq x'$ and both $x \in h_i$ and $x' \in h_i$, then player i, who moves at x, does not know whether she is at x or at x';
3. if $x \neq x'$ and both $x \in h_i$ and $x' \in h_i$, then $A_i(x') = A_i(x)$.

Information sets

The simultaneous-move battle of the sexes: player 2 does not know whether she is at x_1 or at x_2. Two common ways of drawing the information set:



Perfect vs. imperfect information

A game of complete information can have:

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Perfect vs. imperfect information

A game of complete information can have:

- **perfect information:** every information set is a singleton and there are no moves of Nature;
- **imperfect information:** some information sets contain several nodes, or there are moves of Nature.

Strategies

A **pure strategy** for player i is a **complete plan of play**: it describes which pure action player i will choose at each of her information sets.

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A **pure strategy** for player i is a **complete plan of play**: it describes which pure action player i will choose at each of her information sets.

Definition

A pure strategy for player i is a mapping $s_i : H_i \to A_i$ that assigns an action $s_i(h_i) \in A_i(h_i)$ to every information set $h_i \in H_i$. We denote by S_i the set of all pure strategies of player i.

Example 1

For the sequential and the simultaneous-move battle of the sexes, write down:

- the set of players;
- the set of nodes;
- the set of actions of each player;
- the set of terminal nodes;
- the payoffs;
- the information sets;
- the strategies of each player.

Mixed and behavioral strategies

Definition (mixed strategy)

A **mixed strategy** for player i is a probability distribution over her pure strategies $s_i \in S_i$.

This takes away some of the dynamic flavor: the player *selects a plan at random before the game is played* and then follows that pure strategy.

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Definition (behavioral strategy)

A **behavioral strategy** specifies, for each information set $h_i \in H_i$, an independent probability distribution over $A_i(h_i)$, denoted $\sigma_i : H_i \to \Delta A_i(h_i)$.

With a behavioral strategy, a player *mixes among her actions whenever she is called to play*.

Perfect recall

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For games of perfect recall, Kuhn (1953) proved that mixed and behavioral strategies are equivalent: given the opponents' strategies, the same distribution over outcomes can be generated by either a mixed or a behavioral strategy of player i.

Sequential rationality

Finding a solution...

We would expect rational players to play optimally, given their beliefs, *whenever* they are called to move.

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Finding a solution...

We would expect rational players to play optimally, given their beliefs, *whenever* they are called to move.

Definition (sequential rationality)

Given the opponents' strategies σ_{-i}, we say that σ_i is **sequentially rational** if and only if player i plays a best response to σ_{-i} at each of her information sets.

Backward induction

Proposition

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Any finite game of perfect information has at least one sequentially rational Nash equilibrium in pure strategies. Moreover, if no player gets the same payoff at two different terminal nodes, the game has a unique sequentially rational Nash equilibrium.

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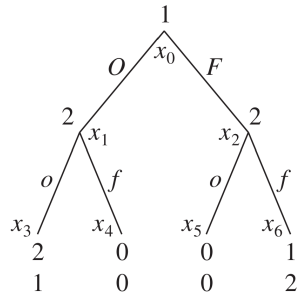
Corollary

Any finite game of perfect information has at least one sequentially rational Nash equilibrium in pure strategies. Moreover, if no player gets the same payoff at two different terminal nodes, the game has a unique sequentially rational Nash equilibrium.

Lesson: "Look forward; reason backward."

Example 2

Solve the sequential battle of the sexes by backward induction.



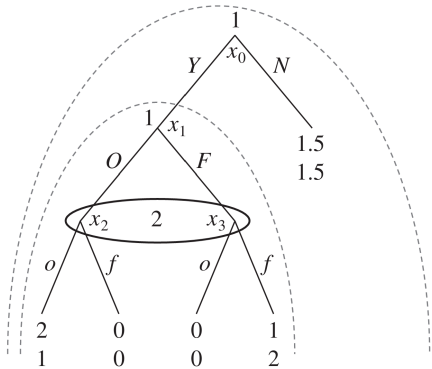
Compare this equilibrium with the equilibria of the normal-form representation of the game.

Subgames

Definition (subgame)

A **proper subgame** G of an extensive-form game Γ consists of a single node and all its successors in Γ, with the property that if $x \in G$ and $x' \in h(x)$, then $x' \in G$. The subgame G is itself a game tree, with information sets and payoffs inherited from Γ.

Subgames



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Subgame-perfect equilibrium

Definition (subgame-perfect equilibrium)

Let Γ be an n-player extensive-form game. A behavioral strategy profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a **subgame-perfect (Nash) equilibrium** if, for every proper subgame G of Γ, the restriction of σ^* to G is a Nash equilibrium in G.

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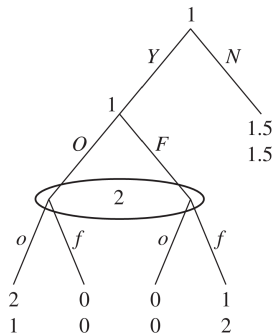
Every subgame-perfect equilibrium is a Nash equilibrium. However, not all Nash equilibria are subgame perfect: subgame-perfect equilibrium refines the set of Nash equilibria, yielding sharper predictions about behavior.

Backward induction vs. subgame-perfect equilibrium

Fact: for any finite game of perfect information, the set of subgame-perfect equilibria coincides with the set of Nash equilibria that survive backward induction.

Example 3

Find all Nash equilibria and all subgame-perfect equilibria of the *voluntary* battle of the sexes:



Example 4: Stackelberg competition

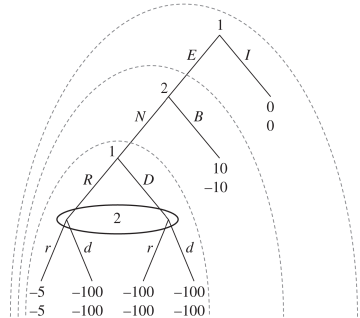
- Two firms, 1 and 2, produce a homogeneous good and choose outputs $q_i \geq 0$.
- Firm 1 moves first: it chooses q_1, and firm 2 observes q_1 before choosing q_2.
- Market demand is linear: $P(q_1 + q_2) = 100 - q_1 - q_2$.
- Cost: $C(q_i) = 10\,q_i$ for both firms.

Find the subgame-perfect equilibrium. How does it compare with the Cournot equilibrium?

Example 5: the Cuban missile crisis of 1962

Player 1 chooses to either ignore the incident (I) or escalate the situation (E). After escalation, player 2 can back down (B) or proceed to a nuclear confrontation (N). Then the players play a simultaneous-move game in which they either retreat (R for player 1, r for player 2) or choose Doomsday (D for player 1, d for player 2).

Find the Nash and the subgame-perfect equilibria.



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Application: monitoring remote work

Motivation: monitoring remote work

- In September 2025, Itaú dismissed employees after finding mismatches between their clock-in records and the activity monitored on their computers while working from home. The bank workers' union estimated about 1,000 dismissals.

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Motivation: monitoring remote work

- In September 2025, Itaú dismissed employees after finding mismatches between their clock-in records and the activity monitored on their computers while working from home. The bank workers' union estimated about 1,000 dismissals.
- Monitoring software is costly for firms, and employees know it may be in place.
- **Question:** how much should a firm monitor, and how much shirking should it expect?

Source: g1 (2025): Itaú demite funcionários após avaliar produtividade no home office

Mini case: work or shirk?

A remote employee chooses to **work** (W) or **shirk** (S); at the same time, the manager chooses to **monitor** (M) or **not** (N). A shirker who is caught loses the wage.

		Manager: M	Manager: N
Employee	W	$w-g,\ v-w-h$	$w-g,\ v-w$
	S	$0,\ -h$	$w,\ -w$

Parameters

$w = 10$	wage
$g = 2$	cost of effort
$v = 20$	value of output
$h = 1$	cost of monitoring

Mini case: questions (in pairs, hand in today)

(a) Show that the game has no Nash equilibrium in pure strategies.

(b) Find the mixed-strategy Nash equilibrium: the probability of shirking and the probability of monitoring.

(c) The firm buys cheaper monitoring software, and h falls from 1 to 0.5. What happens to shirking? And to monitoring? Explain the intuition.

(d) The firm raises the wage to $w = 20$. What happens to shirking and to monitoring?

(e) Write the equilibrium probabilities as functions of w, g and h. Which policy would you recommend to the firm, and which assumptions of the model would you question in the Itaú case?

Wrap-up and next lecture

Today

- Mixed strategies, the principle of indifference and the existence of equilibrium.
- Extensive-form games: game trees, information sets, perfect and imperfect information.
- Pure, mixed and behavioral strategies; perfect recall.
- Sequential rationality, backward induction and subgame-perfect equilibrium.

Next lecture

- Multistage games and bargaining.

Appendix: price competition with capacity constraints

Appendix: price competition with capacity constraints

- Two firms compete simultaneously in prices.
- For simplicity, demand is inelastic and normalized to 1 whenever the price does not exceed the reservation value (also 1).
- Short-run constraints prevent each firm from serving the whole demand. Cost function:

$$c_i(q_i) = \begin{cases} 0 & \text{if } q_i \leq \bar{q}, \\ +\infty & \text{if } q_i > \bar{q}, \end{cases} \qquad \text{with } \bar{q} \in \left(\tfrac{1}{2}, 1\right).$$

Appendix: who serves the demand

Suppose $p_i < p_j$ (firm i serves the demand first):

- if $p_i > 1$, both firms have zero demand;
- if $p_i \leq 1$, firm i serves $\bar{q}$.

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Suppose $p_j < p_i$ (firm j serves the demand first):

- if $p_i > 1$, firm i has zero demand;
- if $p_j < p_i \leq 1$, firm j serves $\bar{q}$ and firm i serves the residual demand $1 - \bar{q}$.

Appendix: payoffs

Payoff of firm i:

$$u_i(p_i, p_j) = \begin{cases} \bar{q}\, p_i & \text{if } p_i < p_j \text{ and } p_i \leq 1, \\ \frac{1}{2} p_i & \text{if } p_i = p_j \leq 1, \\ (1 - \bar{q})\, p_i & \text{if } p_j < p_i \leq 1, \\ 0 & \text{if } p_i > 1. \end{cases}$$

Appendix: best response

Given $p_j \leq 1$, firm i compares two options:

- serve the residual demand $1 - \bar{q}$ at the highest price, $p_i = 1$: profit $1 - \bar{q}$;
- undercut, $p_i = p_j - \varepsilon$, and serve $\bar{q}$: profit close to $\bar{q}\, p_j$.

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If $\bar{q}\, p_j \leq 1 - \bar{q}$, charging 1 is better. If $\bar{q}\, p_j > 1 - \bar{q}$, firm i wants to undercut, but there is no best price just below p_j (indeterminacy). Hence

$$p_i^*(p_j) = \begin{cases} 1 & \text{if } p_j \leq \dfrac{1-\bar{q}}{\bar{q}}, \\ \emptyset & \text{if } p_j > \dfrac{1-\bar{q}}{\bar{q}}. \end{cases}$$

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No pure-strategy equilibrium: against a low p_j, i charges 1; against 1, the rival's best response is empty, since $1 > \frac{1-\bar{q}}{\bar{q}}$.

Appendix: a mixed-strategy equilibrium

Consider a symmetric mixed-strategy equilibrium in which each firm's price has a continuous distribution function F with support $[\underline{p}, \bar{p}]$. One can show that $\bar{p} = 1$.

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By the principle of indifference, $u_i(p_i, F) = u_i(1, F) = 1 - \bar{q}$ for every $p_i \in [\underline{p}, 1]$. Since $\Pr(p_i < p_j) = 1 - F(p_i)$,

$$\left(1 - F(p_i)\right) \bar{q}\, p_i + F(p_i)\,(1 - \bar{q})\, p_i = 1 - \bar{q} \implies F(p_i) = \frac{1}{2\bar{q} - 1}\left(\bar{q} - \frac{1 - \bar{q}}{p_i}\right).$$

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The lower bound solves $F(\underline{p}) = 0$: $\underline{p} = \dfrac{1 - \bar{q}}{\bar{q}}$.

Firms randomize their prices between $\underline{p}$ and 1: a model of price dispersion, as in sales and promotions.

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